JACK M. EWING

Worcester, MA | (737) 600-1915 | jackewing772@gmail.com | linkedin.com/in/Jack-M-Ewing

EDUCATION

Worcester Polytechnic Institute — *B.S. Electrical & Computer Engineering GPA 3.66/4* **Aug 2026 – Expected May 2028**

Lafayette College — *Electrical & Computer Engineering GPA 3.4 / 4* **Aug 2024 – May 2026**

Relevant Coursework: Embedded Design, FPGA Design, Digital Systems, Signals & Systems, Communications, Circuit Analysis, Robotics

TECHNICAL SKILLS

Programming: C, C++, Python, Verilog, SystemVerilog, Java, C#, JavaScript

Embedded & Hardware: ESP32, Arduino, Raspberry Pi, Jetson Nano, FPGA (Artix-7 / Basys 3), I²C, BLE, GPS, IMU, PCB Design, Sensors, Data Acquisition, Soldering, Circuit Debugging

Engineering Tools: KiCad, LTspice, Vivado, SolidWorks, Oscilloscope, Bench Power Supply, Git/GitHub, 3D Printing, Laser Cutting

ENGINEERING EXPERIENCE

Formula SAE — Lafayette College — *Embedded Systems Engineer, Telemetry & Controls* **Jan 2025 – May 2026**

- Designed, built, and tested an ESP32 Feather V2 telemetry system integrating GPS and a BNO055 IMU, capturing live position, acceleration, and orientation data at up to 15 Hz with onboard SD card logging.
- Implemented BLE and I²C communication between embedded boards to stream live vehicle data to a remote tracker; measured approximately 80 ft of reliable line-of-sight BLE range during testing.
- Designed three 2-layer PCBs in KiCad — the I/O board, the connector ("traces") board, and a combined VCU/MPR + 5 V board that consolidated two functions onto one board to reduce total board count; assembled, tested, and debugged the resulting vehicle electronics.
- Designed and tested a five-board modular safety system that commanded the ECU to shut down vehicle systems whenever the safety loop was interrupted; performed system-level integration testing across all boards.
- Debugged the 300 V high-voltage battery charging circuit using an oscilloscope, LTspice simulation, and a bench supply; worked with the Orion BMS to verify charging behavior and safe battery operation.
- Validated the telemetry system on the vehicle during driving tests, collecting clean multi-sensor data streams.

Formula SAE — WPI — *Embedded Systems Engineer, Telemetry & Controls* **Aug 26 - Current**

- Dashboard PCB and waterproofing, The dashboard has CAN connections to the rest of the car, E-fuse, lights and voltage regulator

Brain-Computer Interface Research — Lafayette College — *Undergraduate Researcher* **Sep 2024 – May 2026**

- Collected neural-signal datasets using OpenBCI and commercial BCI hardware for wheelchair-control research, and developed Python workflows for organizing and processing the recorded data.

Simpler Trading — *Software Engineering Intern* **Summer 2026**

- Built a Python tool to combine and evaluate outputs from multiple trading indicators, processing S&P 500 data from 2020–2026 via the Polygon API and generating CSV datasets and analytical visualizations.

PROJECTS